

Outbound Sequencer

Multichannel sequencing engine with deliverability guardrails, reply detection, and automatic pause when a meeting is booked.

TypeScript	Temporal	Postgres	SendGrid
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VERSION	STATUS	CHANNELS	RUNTIME	OWNER	UPDATED
1.9.4	Live	Email, call, task	Node 20	Sales Eng	Sep 2026

1. Overview

The Outbound Sequencer runs multi-step, multichannel outreach on behalf of reps: email steps send automatically, call and social steps land as tasks in the rep's queue. Each enrollment is a durable Temporal workflow, so a sequence spanning six weeks survives deploys, restarts, and outages without losing its place or double-sending a step.

The engine is opinionated about stopping. Sequences pause on any signal that a human has engaged — a genuine reply, a booked meeting, an opportunity created — because the cost of one message sent after a prospect has already answered is far higher than the cost of a missed touch. Deliverability guardrails apply the same logic at the domain level, throttling volume before reputation damage occurs rather than reacting to it.

2. Enrollment lifecycle

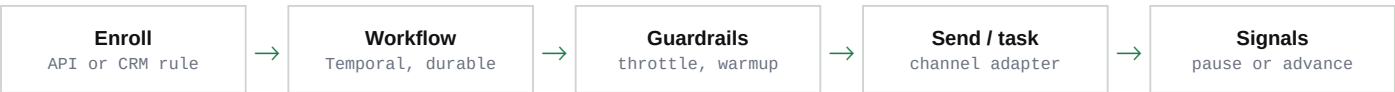


Figure 1 — One workflow per enrollment. Signals arrive asynchronously and can halt the workflow between any two steps.

Sends are scheduled inside the recipient's business hours using the account's timezone, with jitter so a hundred enrollments on the same step do not fire in the same second. A step that cannot send within its window — quota exhausted, domain throttled, holiday calendar — slips to the next window rather than sending late at night or being skipped.

3. Components

Enrollment API	Validates eligibility, checks suppression, and starts one workflow per prospect.
Workflow engine	Temporal workflows hold step state, timers, and retries; activities do the side effects.
Channel adapters	SendGrid for email with per-mailbox identity; call and social steps become CRM tasks.
Guardrail service	Per-domain and per-mailbox rate limits, warmup ramps, bounce and complaint circuit breakers.
Signal listeners	Reply, out-of-office, meeting-booked, and opportunity-created events, normalised to one bus.
Suppression store	Unsubscribes, competitor and customer domains, active-opportunity accounts, do-not-contact.

4. Sequence definition

```
export const sequence = defineSequence({
  name: "enterprise-outbound-q4",
  window: { days: ["Mon","Tue","Wed","Thu"], hours: [8, 16], tz: "prospect" },
  guardrails: { perMailboxDaily: 60, perDomainDaily: 8, warmupDays: 21 },
  pauseOn: ["reply.human", "meeting.booked", "opportunity.created"],
  steps: [
    { day: 0, channel: "email", template: "intro-v4" },
    { day: 2, channel: "call", task: "First attempt, reference intro email" },
    { day: 5, channel: "email", template: "case-study", thread: true },
    { day: 9, channel: "social", task: "Connection request, no pitch" },
    { day: 14, channel: "email", template: "breakup", thread: true },
  ],
  onExhausted: "return-to-nurture",
});
```

Listing 1 — Step timing is relative to enrollment; the engine resolves it against the send window.

5. Reply detection and auto-pause

- **Classification.** Inbound mail is classified as human reply, auto-reply, out-of-office, or bounce. Only a human reply pauses the sequence; an out-of-office defers the next step until the stated return date.
- **Meeting signals.** A calendar invite accepted by the prospect, or a meeting logged against the account in the CRM, pauses every active enrollment for that account — not just the one that produced the meeting.
- **Account-level safety.** When an opportunity is open, all prospecting enrollments at that account suspend automatically, which prevents cold outreach from reaching a live deal's buying committee.

6. Deliverability guardrails

CONTROL	DEFAULT	TRIP BEHAVIOUR
Per-mailbox volume	60 sends/day after warmup	Excess slips to the next send window.
Warmup ramp	5/day, +5 daily for 21 days	New mailboxes cannot skip the ramp.
Bounce rate	2% over 500 sends	Mailbox paused; owner and admin notified.
Complaint rate	0.1% over 1,000 sends	Domain-wide halt pending manual review.

7. Operations

- Temporal history is the audit log: for any prospect, the exact sequence of sends, skips, signals, and pauses is reconstructible, which matters when someone asks why a message went out.
- Template and guardrail changes apply to in-flight enrollments at the next step boundary; already-scheduled sends are not rewritten retroactively.
- Suppression is checked twice — at enrollment and again immediately before each send — so a mid-sequence unsubscribe cannot be raced by an already-scheduled step.

8. Roadmap

RELEASE	PLANNED WORK	TARGET
2.0	Per-prospect send-time optimisation from historical open and reply timing.	Q4 2026
2.1	Native LinkedIn and SMS adapters behind the existing channel interface.	Q1 2027
2.2	Sequence A/B testing with significance reporting at the step level.	Q2 2027